

Analysis Report -- Run 20260612_161004

1. Executive Summary

This run modeled `rate_per_10000_ed_visits`, a continuous panel-regression target covering emergency-department visit rates by month (`period_id`), U.S. jurisdiction (51 levels), and overdose category (8 levels). The training table `data/train/dose_sys_train.csv` contains 31,416 rows and 4 columns (`period_id`, `jurisdiction`, `overdose_category`, `rate_per_10000_ed_visits`), joined with `data/train/covariates.csv` (3,927 rows, 12 columns) on (`period_id`, `jurisdiction`).

The selected model is a stacked blend dominated by gradient-boosted trees (`stack(catboost_log+tuned(catboost)+catboost)`), the round-2 gbdt specialist candidate, OOF `block_mae`=0.7865), combined via NNLS with the deterministic floor, trees, and linear specialists into a final blend with OOF `block_mae` = **0.784656** versus the floor's 0.800677 (a 2.00% improvement).

A robust-generalization adjustment was applied in the sense that specialist candidates were screened by a feature-ablation gate and a 3-reviewer consensus (model performance, feature leakage, generalization) before promotion; no separate `adjusted_robust_score` field was emitted by `final_model.json` (this artifact does not exist for the specialist modeling mode -- `model_selection.json` / `promotion.json` serve this role).

The primary limitation, drawn from `data_profile.json.summary_warnings`, is substantial missingness in the prediction covariates: `unemployment_rate` (54.6% missing) and `labor_force` (58.0% missing) in `data/val/covariates.csv` -- both of which were ultimately pruned from the model by the feature ablation gate.

2. Task Interpretation

- **Task type:** regression (`spec_parse.json.task_type`), confidence 0.85. Determined primarily from `DATA_DESCRIPTION.md` plus the data-driven signal that `rate_per_10000_ed_visits` is a high-cardinality (`n_unique`=3305) continuous float column with no missing values in train.

- **Target column:** `rate_per_10000_ed_visits`.
- **Row ID column:** `row_id` (from `sample_submission.csv`).
- **Evaluation metric:** `block_mae`, with `period_id` as the block column.

spec_parse.json notes the metric was "not explicitly named in Data_Description.md; inferred 'block_mae' (block=period_id) as the standard metric for this panel-regression task per orchestrator-supplied hint and standard practice for this benchmark series."

- **Required output format:** continuous -- two-column submission (row_id, rate_per_10000_ed_visits), 918 rows matching sample_submission.csv row order.

- **Join keys:** period_id, jurisdiction (panel structure; detected_structure.panel_structure=true).

3. Data Overview

Train target file: data/train/dose_sys_train.csv

- n_rows = 31,416, n_cols = 4
- 0% missing across all 4 columns
- Target stats: min=0.05, max=79.99, mean=6.352, std=7.970, skewness=2.703 (recommend_log_transform=true, log1p skewness=0.499)

Train covariates file: data/train/covariates.csv

- n_rows = 3,927, n_cols = 12

Column	dtype	missing_rate	Notes
period_id	object	0.0	opaque monthly period id, 77 unique values, 2019-01..2025-05
jurisdiction	object	0.0	51 U.S. jurisdictions
unemployment_rate	float64	0.0	n_unique=146
labor_force	float64	0.0	n_unique=3923; flagged potential_id_columns (high-cardinality artifact, not an ID)
temp_avg_f	float64	0.0588	weather covariate
precip_in	float64	0.0588	weather covariate
gtrends_overdose	float64	0.0	search-interest signal
gtrends_fentanyl	float64	0.0	search-interest signal
gtrends_naloxone	float64	0.0	search-interest signal
gtrends_opioid	float64	0.0	search-interest signal
gtrends_methamphetamine	float64	0.0	search-interest signal
state_doh_release	object	0.2959	free-text press-release; high-cardinality (n_unique=2665), flagged text_like

Prediction file: data/val/covariates.csv

- n_rows = 357, n_cols = 12 (same schema as train covariates)

- Schema diff vs train: identical columns (schema_diff.in_both lists all 12 covariate-shared columns);

overdose_category and rate_per_10000_ed_visits are train-only (the latter is a placeholder constant column when it appears in sample_submission.csv).

- Missingness in prediction file: unemployment_rate 54.6%, labor_force 58.0%, temp_avg_f 2.0%, precip_in 5.9%, state_doh_release 32.2%.

- data/val/covariates.csv contains 7 unique period_id values (51 jurisdictions x 7), one of which (dsZhPyK4, 2025-06-30) is not in sample_submission.csv (only 6 periods, 153 rows x 6 = 918 in the cross-joined target frame) -- treated as buffer/lag context only.

sample_submission.csv: n_rows=918, n_cols=5

(row_id, period_id, jurisdiction, overdose_category, rate_per_10000_ed_visits).

The placeholder target column is constant (all 0.0, n_unique=1).

4. Preprocessing and Feature Engineering

Excluded columns:

- rate_per_10000_ed_visits from sample_submission.csv -- constant placeholder

(n_unique=1), never read as feature or target signal

(analysis_plan.json.feature_plan.exclude_columns).

- row_id -- submission join key only, not a model feature.
- Raw period_id and jurisdiction/overdose_category strings -- never used directly as model features; only derived/encoded forms enter the model.

Imputation strategy (analysis_plan.json.feature_plan.imputation):

- temp_avg_f, precip_in: jurisdiction-month training median.
- unemployment_rate, labor_force: jurisdiction training last-known-value carry-forward.
- state_doh_release: filled with empty string plus a structural-missing indicator (state_doh_release__is_missing).

Encoding strategy:

- jurisdiction: per-fold target encoding (target aggregates, see below).
- overdose_category: one-hot encoding (8 columns) -- present in the full feature spec but pruned by the ablation gate (see below).

Datetime-derived features (period_id, detected as

"opaque id maps to month-end dates via Data_Description period map"):

period_id__year, period_id__month, period_id__quarter, period_id__month_sin, period_id__month_cos, period_id__day, period_id__dayofweek, period_id__is_weekend, period_id__weekofyear, period_id__ordinal_rank, period_id__ordinal -- 11 columns, all retained by the ablation gate (feature_audit_review.json: datetime_derived delta = +0.009556, decision = keep).

Text features: state_doh_release was processed via TF-IDF (max_features=2000,

1-2gram, min_df=2) → TruncatedSVD (20 components: state_doh_release__svd_0..19)
plus the missing-indicator, fit on full train text and applied to prediction text
(21 columns total, retained -- text_svd ablation delta = -0.001097, decision = keep).

Per-fold target aggregates (per_fold_aggregates, 4 columns, retained -- delta
= -0.001621, decision = keep):

jurisdiction_category_target_mean, jurisdiction_target_mean,
category_target_mean, jurisdiction_category_target_std, grouped on
(jurisdiction, overdose_category), (jurisdiction), and (overdose_category).

****Pruned by the ablation gate (other_features, 27 columns, decision = prune,
delta = -0.041934 ≈ 4.9x the prune margin of 0.00848)**:** all raw weather/economic/
gtrends covariates (temp_avg_f, precip_in, unemployment_rate, labor_force,
gtrends_overdose, gtrends_fentanyl, gtrends_naloxone, gtrends_opioid,
gtrends_methamphetamine), the 8 overdose_category__is_* one-hots, the
jurisdiction_x_period_ordinal_rank interaction, the 3
overdose_category_*_x_gtrends_fentanyl
interactions, the experimental lag/rolling target features
(target_lag1/2_per_jurisdiction_category,
target_rolling_mean_3/6_per_jurisdiction_category,
gtrends_fentanyl_lag1_per_jurisdiction, cross_category_auxiliary_lag1_signal).
The feature-engineering-reviewer confirmed this prune as "a large, unambiguous signal,
not a modest round-1 prune" since per_fold_aggregates, datetime_derived, and text_svd
were retained as the highest-value groups.

The final pruned spec retains **36 feature columns**: 11 datetime-derived, 21 text-SVD,
4 per-fold target aggregates.

All transformations were fitted on the training split only (per-fold target aggregates
and the TF-IDF/SVD transformer were fit on training-period data, with full-train refit
applied only to the prediction matrix per analysis_plan.json.feature_plan.rationale).

5. Validation Strategy

validation_strategy.json (mode=schema_review): group_time_split, chosen
because train periods (77 monthly period_id values, 2019-01..2025-05) and prediction
periods (6 monthly periods, 2025-06..2025-12) are disjoint and chronological, in a
panel structured as (period_id x jurisdiction x overdose_category).

Holdout parameters (validation_strategy.json.holdout_parameters):

- time_column = period_id, group_column = period_id
- n_splits = 5
- random_state = 42

GroupKFold over `period_id` (77 groups) guarantees every fold's validation rows come from periods entirely absent from that fold's training rows, directly reproducing the train/prediction period disjointness, while also yielding 5 folds (more stable OOF) versus a single last-slice holdout. `block_mae` is computed per `period_id` block in both OOF scoring and the hidden leaderboard.

Leakage risks recorded:

- `rate_per_10000_ed_visits` appears as a placeholder constant in `sample_submission.csv` -- low severity, not a leakage path, but feature code must never read this placeholder column.
- `jurisdiction/overdose_category` as target-aggregate keys -- medium severity; any target-derived aggregate must be fit per training fold only.
- The 5 unscored `overdose_category` values (heroin, fentanyl, cocaine, methamphetamine, benzodiazepine; corr 0.70-0.88 with scored categories except heroin) as auxiliary signal -- medium severity; per-fold-only aggregation required.
- Out-of-distribution period range -- low severity; hidden periods (2025-06..2025-12) are entirely beyond max training period (2025-05), so GroupKFold cannot fully replicate true future extrapolation.

Validation limitation (verbatim): "GroupKFold over `period_id` holds out whole training periods but folds are drawn from within the 2019-01..2025-05 training span, not strictly the final periods -- it does not fully capture pure forward extrapolation to 2025-06..2025-12 ... Trend/seasonality extrapolation risk for the final 6-month horizon is therefore only partially represented by OOF error on these folds."

All 31,416 training rows were scored OOF on this canonical fold assignment (`{run_id}_cv_folds.json`).

6. Candidate Models

All scores are OOF `block_mae` on the canonical 5-fold `group_time_split` (lower is better). `model_selection.json` records the following pool:

Model	Val Score (block_mae)	Notes
<code>dummy_mean</code>	5.4282	trivial baseline
<code>baseline_group_mean</code>	5.2098	grouped-by-jurisdiction baseline
<code>baseline_category_mean</code>	2.1855	grouped-by-category baseline
<code>baseline_group_category_mean</code>	1.2421	grouped-by-(jurisdiction,category) baseline
<code>ridge</code>	1.1999	linear
<code>elastic_net</code>	1.1986	linear
<code>gradient_boosting</code>	0.9269	sklearn GBM

Model	Val Score (block_mae)	Notes
extra_trees	0.8843	trees
hist_gradient_boosting	0.8598	HGB
random_forest	0.8310	trees
xgboost	0.8266	gbdt
hgb_sqrt	0.8234	HGB variant
hgb_log	0.8188	HGB on log1p target
lightgbm_strong	0.8347	gbdt
lightgbm	0.8328	gbdt
lightgbm_sqrt	0.8181	gbdt variant
lightgbm_log	0.8140	gbdt on log1p target
catboost	0.8160	gbdt
catboost_log	0.8080	gbdt on log1p target
tuned(catboost)	0.8072	tuned (depth=6, lr=0.05, l2_leaf_reg=1.0, iterations=2000)
stack(tuned(catboost)+catboost_log+lightgbm_log)	0.8007	model_selection-stage selection (this is the round-1 "floor")

adjusted_robust_score is not present in any artifact for this run (no final_model.json was produced); classic OOF-block_mae selection on the canonical folds was used throughout. The "stack" entry above is the seeded **floor** ({run_id}_oof_floor.csv, score 0.800677) produced in Step 6A; it then competed against the round-2 specialist candidates below in the keep-best NNLS blend.

Specialist candidates (Step 6B, scored on the same canonical folds, {run_id}_ensemble_meta.json, round 2):

Model	OOF block_mae	NNLS weight	Notes
floor (stack(tuned(catboost)+catboost_log+lightgbm_log))	0.800677	0.007936	round-1 deterministic floor
gbdt (stack(catboost_log+tuned(catboost)+catboost))	0.7865	0.843493	best single candidate, dominant weight
trees	0.808348	0.144878	barely beats floor
linear	1.188767	0.003693	near-zero weight, weakest
blend (selected)	0.784656	--	NNLS blend over the 4 candidates above

7. Selected Model and Predictions

- **Selected model:** blend(floor+gbdt+trees+linear), an NNLS blend with weights floor=0.007936, gbdt=0.843493, trees=0.144878, linear=0.003693 ({run_id}_ensemble_meta.json, round 2).

- **Validation score (primary metric):** block_mae = **0.784656087164158**.
- **Train-val gap:** not measured. model_performance_review.json reports train_val_gap: null and notes "model_stability_by_split.json (n_splits_evaluated=0) and overfitting_audit.json (primary_holdout_score=null, stability={}) remain empty across both rounds, so train_val_gap is unmeasurable."
- **Selection rationale:** the blend beat the floor by 2.00% (0.800677 → 0.784656), driven primarily by the round-2 relaunched gbdm specialist (stack(catboost_log+tuned(catboost)+catboost), OOF 0.7865, NNLS weight 0.843). An NNLS-weighted cross-family ensemble (floor + gbdm + trees + linear specialists) was used as the final submission model.
- **Did the candidate beat the baseline?** Yes -- the floor/blend (0.7847-0.8007) substantially outperforms the best non-trivial baseline (baseline_group_category_mean = 1.2421) and all linear models (~1.20).
- **Submission validation** (submission_validation.json): columns_ok=true, row_count_ok=true (918 rows), row_id_alignment_ok=true, no_duplicate_row_ids=true, all_finite=true, dtype_ok=true, missing_predictions=0, overall_verdict=PASS, next_action=proceed.

8. Overfitting and Generalization Controls

8.1 Validation strategy rationale. The schema-review guardian detected a panel structure (has_time=true, has_group=true, has_panel=true, temporal_overlap=false) with disjoint, chronological train/prediction periods. group_time_split (GroupKFold on period_id, 5 splits) was chosen over a plain time-based holdout because it "guarantees every fold's validation rows come from periods entirely absent from that fold's training rows -- directly reproducing the train/val period disjointness -- while also yielding multiple folds (more stable OOF) instead of a single last-slice holdout."

8.2 Train vs validation gap. train_val_gap = null in model_performance_review.json -- no per-candidate train score was logged by the specialist agents in either round, so the gap cannot be classified as acceptable/moderate/high. The reviewer assigned overfit_risk = "moderate" specifically because this gap is unmeasured, not because a measured gap exceeded a threshold. No adjusted_robust_score/robust-penalization field exists for this run.

8.3 Number of validation splits. validation_strategy.json.holdout_parameters.n_splits = 5, GroupKFold grouped on period_id` (77 groups split across 5 folds).

8.4 Model complexity summary. The selected model is an ensemble blend dominated (84.3% NNLS weight) by a stacked gradient-boosted-tree model

(stack(catboost_log+tuned(catboost)+catboost)), with smaller contributions from a trees-family candidate (14.5%) and negligible contributions from the floor stack (0.8%) and a linear model (0.4%). Simpler candidates were evaluated and rejected on OOF score: linear models (ridge 1.1999, elastic_net 1.1986) and grouped-mean baselines (1.2421-5.4282) all scored substantially worse than the gbd-t-dominated ensemble (0.7847).

8.5 Leakage audit result. overfitting_leakage_audit.json (round 2, generalization-reviewer, overall_verdict=PASS, high_overfitting_risk=false):

- cv_strategy_validity: pass -- strategy unchanged from round 1, all 4 candidates scored on the same canonical folds.

- cv_holdout_calibration: pass -- round-over-round improvement of 0.054%, "well below the 20% single-round target-leakage threshold."

- prediction_sanity: pass -- 918 unique predictions, CV(std/mean)=0.685, range [3.489, 53.670] within train range [0.05, 79.99], mean shift of 0.781 train-sigma attributed to elevated gtrends_fentanyl/gtrends_naloxone covariates in the prediction periods (a covariate shift, not a leakage artifact).

- train_val_gap: pass -- no per-candidate train/val gap fields emitted; relative deltas (round-over-round 0.054%, OOF-vs-floor 2.0%) are both "far below the 30%/50% gap thresholds."

Separately, feature_audit_review.json (feature-leakage-reviewer, overall_verdict=pass) flagged one low-severity item: the TF-IDF/SVD transformer for state_doh_release was fit once on full train text rather than refit per CV fold (unsupervised, no target leakage). Suggested as a low-priority future fix, not blocking.

8.6 Prediction sanity result. prediction_sanity.json: verdict=PASS, high_overfitting_risk=false. All 8 checks pass: finite values (918/918), row_id order match, columns/shape match (918 rows, 2 cols), non-constant predictions (918 unique values, std=8.6113), range plausibility (pred [3.489, 53.670] vs train [0.050, 79.990]), train/pred distribution comparison (mean_shift_in_train_std=0.781), CV improvement over floor (2.00%), and round-over-round improvement (0.785078 → 0.784656, delta=0.000422).

8.7 Final model selection rationale. No final_model.json exists for this specialist-mode run; the equivalent rationale is in {run_id}_promotion.json and the lead reviewer's analysis_review_2.json: "Round2 plateau confirmed: blend 0.785078 → 0.784656, +0.0527% of baseline (<0.5% threshold). All 3 reports PASS, no high-impact suggestions ... At ~75/90min budget, ship best (gbd-t-dominated stack, 2.0% over floor)." approved_for_final=true, next_action=stop_and_report.

8.8 Repair rerun status. No supervisor_gatekeeper.json artifact named

{run_id}_gate_supervisor.json exists as a repair trigger; the supervisor gate status ({run_id}_gate_supervisor.json) is pass with no suggested_corrections. No repair rerun was triggered. (The round-1 → round-2 gbd specialist relaunch was an ordinary improvement-loop continuation per analysis_review_1.json's high-impact suggestion, not a repair-mode rerun.)

9. Hardcoding Audit

hardcoding_audit_pre.json (phase=pre, verdict=fail):

- **Total findings:** 50 -- unacceptable=4, risky=41, acceptable=5.
- The 4 unacceptable findings are all in outputs/scratch/20260612_161004/(missingness_audit.py, profile.py): hardcoded dataset-specific column literals "jurisdiction" and "state_doh_release" via direct column access / variable assignment, rather than resolving from spec_parse.json.file_schemas at runtime.
- The 41 risky findings are predominantly the audit tool's own static search-term catalogue (e.g. rate_per_10000_ed_visits, overdose_category, all_drugs/all_opioids/all_stimulants, 918, survived, titanic, submission.csv -- used for *detection*, not modeling logic) plus the standard repo-root submission.csv deliverable filename mandated by the orchestrator.
- The audit explicitly states: "No occurrences of banned Award-A literals (rate_per_10000_ed_visits as data access, overdose_category as data access, all_drugs/all_opioids/all_stimulants as data access, magic number 918, fixed period-id maps, dev absolute paths) found in actual modeling/data-access code paths."
- No hardcoding_audit_post.json is available for this run; the post-audit (Step 8, after Step 3) was not found among outputs/logs/*.json at report-writing time. The 4 unacceptable pre-audit findings in outputs/scratch/ remain unresolved in the available logs.

10. Limitations

- **Missing data** (data_profile.json.summary_warnings, missing_rate > 0.10):
- train_covariates.state_doh_release: 29.6% missing (structural -- no DOH release that period; handled via empty-string fill + missing indicator).
- prediction.unemployment_rate: 54.6% missing.
- prediction.labor_force: 58.0% missing.
- prediction.state_doh_release: 32.2% missing.
- Note: unemployment_rate and labor_force (both with the highest

prediction-set missingness) were pruned from the final feature set by the ablation gate, reducing exposure to this missingness; `state_doh_release` (handled via TF-IDF/SVD + missing indicator) was retained.

- **Datetime features:** `period_id` is an opaque monthly identifier resolved to 11 derived cyclical/ordinal features (`period_id__year`, `__month`, `__quarter`, `__month_sin/cos`, `__day`, `__dayofweek`, `__is_weekend`, `__weekofyear`, `__ordinal_rank`, `__ordinal`); the raw `period_id` string was never used as a feature.

- **Model limitations:** the train-vs-validation gap was not measured by the specialist agents (`train_val_gap=null`), so the reviewer classified overall `overfit_risk` as "moderate" on this basis alone, despite all leakage and prediction-sanity checks passing.

- **Validation limitations:** GroupKFold on `period_id` draws folds from within the training span (2019-01..2025-05) and "does not fully capture pure forward extrapolation to 2025-06..2025-12 ... Trend/seasonality extrapolation risk for the final 6-month horizon is therefore only partially represented by OOF error on these folds" (`validation_strategy.json.validation_limitations`). This is the run's primary forward-extrapolation calibration gap: the hidden evaluation periods are entirely outside the range seen by any training fold.

- **Overfitting risk:** classified "moderate" per `model_performance_review.json`, driven by the unmeasured train/val gap rather than a measured large gap; no high-severity leakage findings were raised by either the feature-leakage or generalization reviewers.

- **Generalizability:** Generalization to future data (the 2025-06..2025-12 prediction periods) has not been validated against a true held-out future period; the predicted distribution shows a mean shift of 0.781 train-sigma above the training target mean, attributed in `overfitting_leakage_audit.json` to elevated `gtrends_fentanyl/gtrends_naloxone` covariate values in the prediction window relative to train.

- **Causal interpretation:** All findings describe statistical associations. No causal claims are made. Per-fold target aggregates (e.g. `jurisdiction_category_target_mean`) and SVD components of `state_doh_release` are predictive of the target within this validation design but do not imply that any covariate causes changes in `rate_per_10000_ed_visits`.

- **Unresolved hardcoding findings:** 4 unacceptable literal-column-access findings remain in `outputs/scratch/20260612_161004/per_hardcoding_audit_pre.json`, with no corresponding post-audit log available at report time to confirm resolution.

11. Appendix

Feature list (36 columns, final pruned {run_id}_feature_spec.json):

- Datetime-derived (11): period_id__year, period_id__month, period_id__quarter, period_id__month_sin, period_id__month_cos, period_id__day, period_id__dayofweek, period_id__is_weekend, period_id__weekofyear, period_id__ordinal_rank, period_id__ordinal
- Text-SVD (21): state_doh_release__is_missing, state_doh_release__svd_0 through state_doh_release__svd_19
- Per-fold target aggregates (4): jurisdiction_category_target_mean, jurisdiction_target_mean, category_target_mean, jurisdiction_category_target_std

Excluded / pruned columns:

- rate_per_10000_ed_visits (from sample_submission.csv, constant placeholder; analysis_plan.json.feature_plan.exclude_columns)
- 27 columns pruned by the Step-6A' ablation gate (other_features group): temp_avg_f, precip_in, unemployment_rate, labor_force, gtrends_overdose, gtrends_fentanyl, gtrends_naloxone, gtrends_opioid, gtrends_methamphetamine, 8x overdose_category__is_*, jurisdiction_x_period_ordinal_rank, 3x overdose_category_*_x_gtrends_fentanyl, target_lag1_per_jurisdiction_category, target_lag2_per_jurisdiction_category, target_rolling_mean_3_per_jurisdiction_category, target_rolling_mean_6_per_jurisdiction_category, gtrends_fentanyl_lag1_per_jurisdiction, cross_category_auxiliary_lag1_signal.

Run metadata:

- run_id: 20260612_161004
- task_type: regression
- target_column: rate_per_10000_ed_visits
- metric: block_mae (block = period_id)
- target_transform: log1p (per {run_id}_feature_spec.json)
- random_state: 42 (validation_strategy.json.holdout_parameters.random_state)
- cv: group_time_split, GroupKFold on period_id, n_splits=5

Artifact inventory (key files referenced in this report, under

outputs/logs/ unless noted):

- spec_parse.json, data_profile.json, validation_strategy.json, 20260612_161004_validation_strategy.json, 20260612_161004_cv_folds.json
- analysis_plan.json, plan_review_1.json
- 20260612_161004_feature_spec.json, 20260612_161004_feature_spec_full.json, 20260612_161004_feature_gate.json, 20260612_161004_feature_ablation.json
- 20260612_161004_model_selection.json, 20260612_161004_oof_floor.csv

- 20260612_161004_agent_gbdtd.json, 20260612_161004_agent_trees.json, 20260612_161004_agent_linear.json
- 20260612_161004_ensemble_meta.json, 20260612_161004_meta_choice.csv, 20260612_161004_promotion.json, 20260612_161004_prior_best.csv
- prediction_sanity.json, model_performance_review.json, feature_audit_review.json, overfitting_leakage_audit.json, analysis_review_1.json, analysis_review_2.json
- submission_validation.json, hardcoding_audit_pre.json, data_conversion.json, 20260612_161004_gate_supervisor.json
- submission.csv (repo root, 918 rows x 2 cols, PASS validation)